

Robots in Context

Programming Exercise 2

Leon Bodenhagen lebo@mmmi.sdu.dk

Associate Professor

SDU Robotics, Maersk Mc-Kinney Moller Institute



Representation of the workspace

- Images
 - White represents free space, black represents occupied space
- You can use other colors to mark the paths



Programming Exercise - Task

- Implement a program which can
 - Read in an image as workspace (**Not allowed to use Nav2**)
 - Can plan a path between two points in the image using at least one of the Bug algorithm, ideally two.
 - Can output the results, e.g., superimposed on the image(s) in different colors.
- Test the program on different maps with different complexity.



Programming Exercise 2 - Reporting

- General: 7ish minute presentation
 - Explain how you wanted to solve the problem; which bug algorithm(s), etc.
 - Implementation: Explain how you solved it, i.e. what tools (e.g. Matlab) you used
 - Evaluation: outline your evaluation strategy and show results. Consider potentially how to assess the performance and how to report it.
 - Again: Mention also challenges
 - Most likely someone else in this room will face the same problems
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ROS2 – Quick Guide

Load Map

Have a look at the OccupancyGrid map package in ROS2

Don't use Nav2

RVIZ

- Use RVIZ to visualize the paths generated by the bug algorithms.

- You can use 2D Pose Estimate to define start pos, and 2D Goal Pose to define the goal

Draw line in RVIZ

As a help to actually draw the bug path, check:

<https://www.geeksforgeeks.org/dsa/bresenham-line-generation-algorithm/>

